IT Applications in Business

Service Level Management, Availability and Continuity Management

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HR EXCELLENCE IN RESEARCH

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Lecture 2. ***Service Level Management, Availability and Continuity Management***

1. ***Service Level Management***
   - Purpose and application of service level management
   - Scope of SLM
   - Value for business
   - Activities related to the process
   - Types of contracts (OLA, UN, SLA)
2. ***Availability Management***
   - Availability Management essentials
   - Definitions: *Availability / Reliability / Maintainability / Serviceability*
   - The calculation of service availability
   - Availability expressed in nines

Lecture 2. ***Service Level Management, Availability and Continuity Management***

3. ***Availability Management***
   - Accessibility management, and other ITIL processes
   - Cost of unavailability of services
4. ***Capacity Management*** and ***Service Continuity Management***
   - Introduction to IT Service Continuity Management (ITSCM)
   - Restoring services to operation (Cold, Warm, Hot reserves)
5. ***Security vs. Safety***
6. ***ITIL v3 vs. v4 – essential differences***

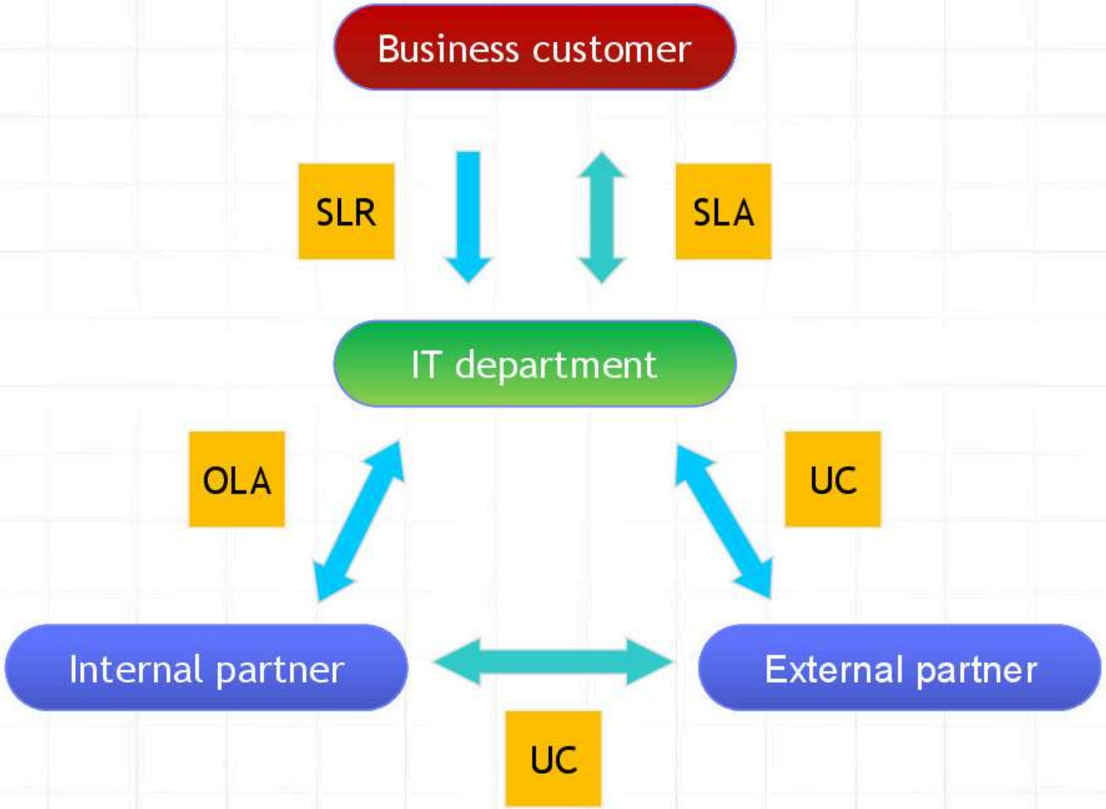
Service Level Management

- Process aim: agree and ensure the terms of providing a specific service to a specific recipient
  - ***Service Level Requirements (SLRs)*** – a document containing customer requirements for the services they want to use
  - ***Service Specification*** - service specification, translation of customer requirements to “how" the supplier will provide them
  - ***Service Level Agreement (SLA)*** - a document setting agreed service levels between a customer and a service provider

Service Level Management

- ***Underpinning Contract (UC)***
  - a document defining agreed service levels between an organization and an <u>external</u> supplier / vendor
- ***Operational Level Agreement (OLA)***
  - a document setting agreed service levels between an organization and another <u>internal</u> provider (e.g. another department)
- ***Service Quality Plan (SQP)***
  - contains information about the <u>*key performance indicators*</u> of the IT organization in the field of service measurement

Service Level Management



Availability Management

- <u>Process aim</u> maintain* levels of services provided, written in SLAs, in order to meet the current and future (agreed) needs of the enterprise in a cost-effective manner

* Related to availability of services and configuration elements - <u>not people</u>.

Availability Management

- ***Availability***
  - the ability of a service, or a configuration element to perform the required function at a given time or period, expressed as a percentage:

$$\boldsymbol{Availability} = \frac{Agreed\ Service\ Time\ - Downtime}{Agreed\ Service\ Time}\ x\ 100\%$$

Availability Management

- ***Reliability***
  - a measure of how long the service or configuration element can perform the agreed upon functions without interruption

Mean Time Between Failures (MTBF) - expressed in hours

$$MTBF = \frac{Available\ time\ - Total\ downtime}{Number\ of\ breaks}$$

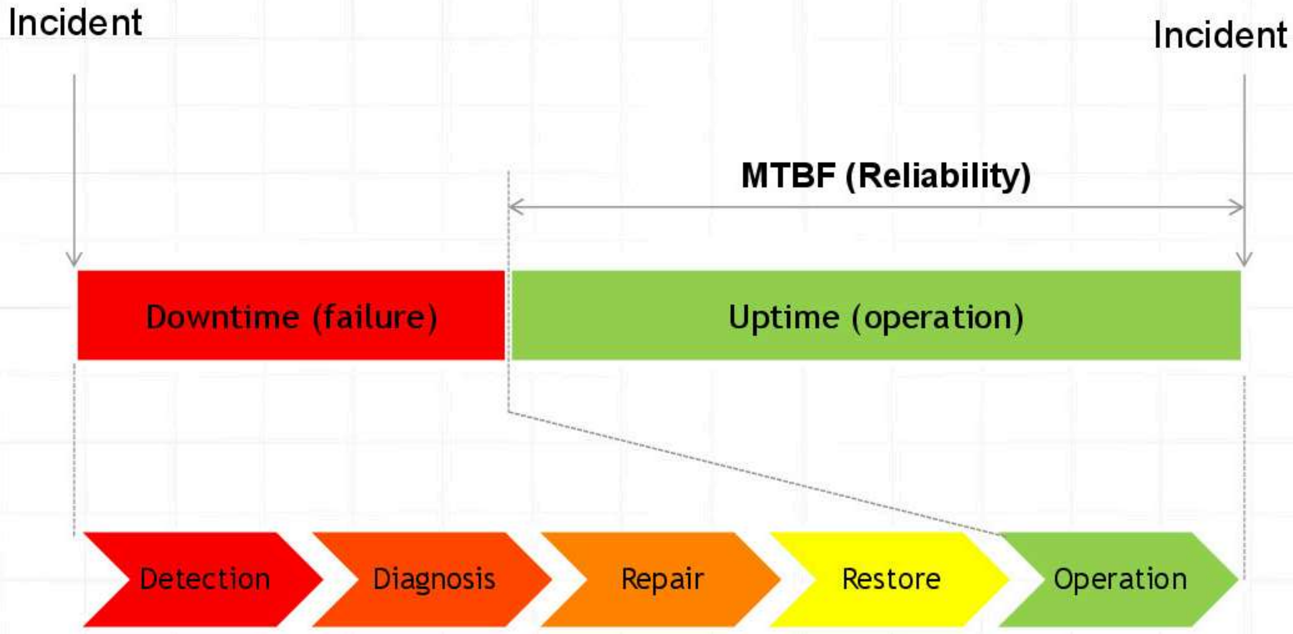
Mean Time Between Service Incidents (MTBSI) - expressed in hours

$$MTBSI = \frac{Available\ time}{Number\ of\ breaks}$$

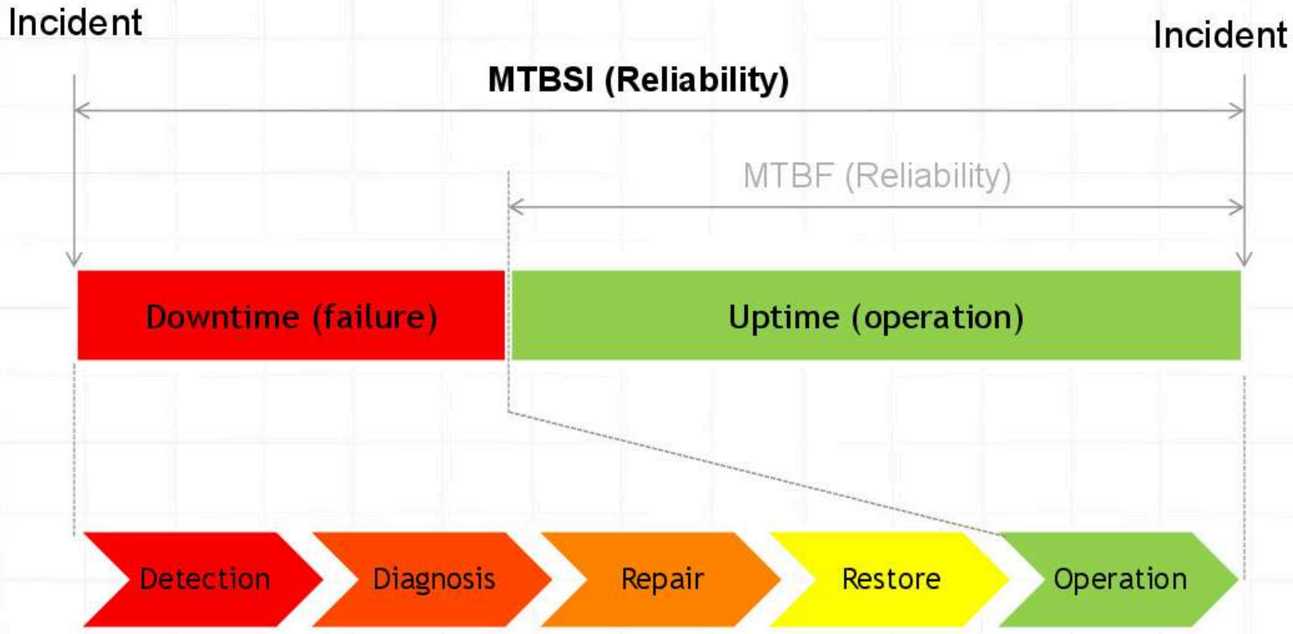
Availability Management

- Reliability of the service can be improved by increasing :
  - reliability of individual elements
  - service resilience to a failure of individual elements

Availability Management



Availability Management

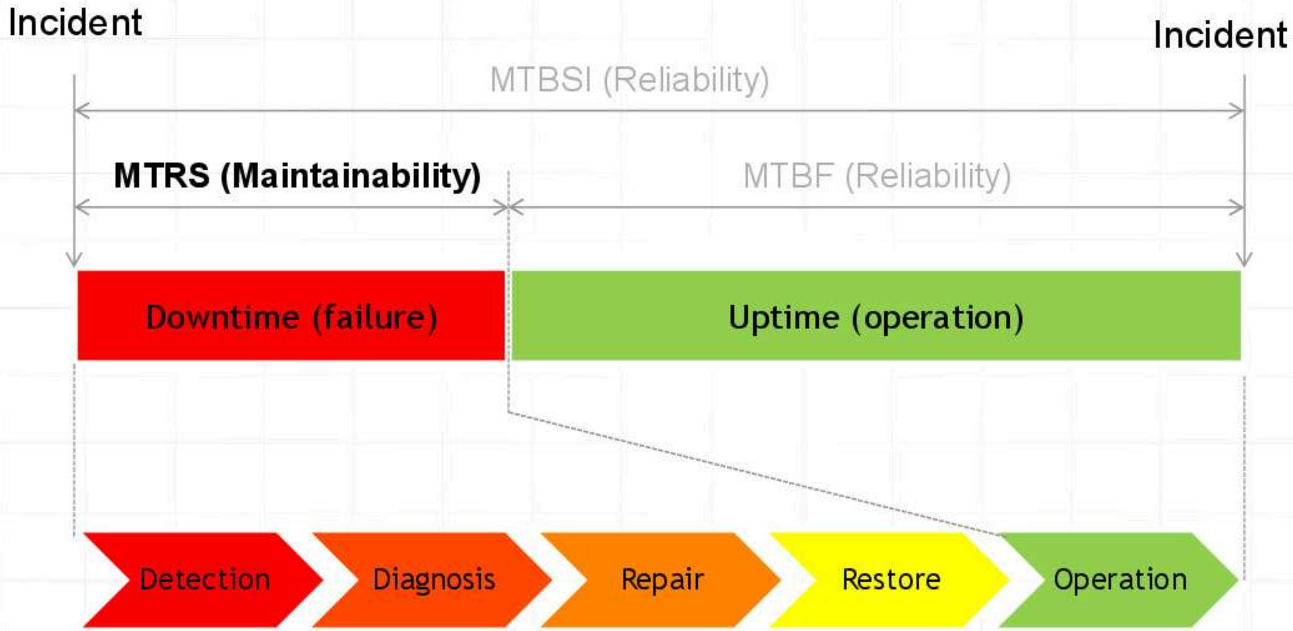


Availability Management

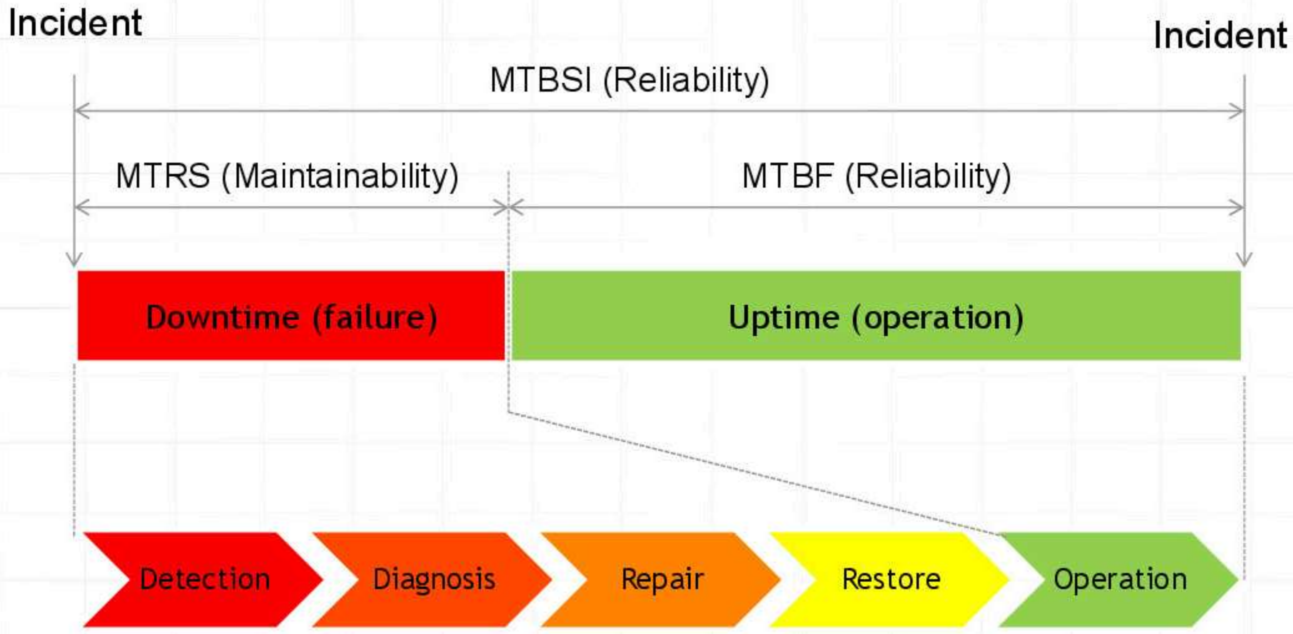
- ***Maintainability***
  - a measure of how quickly and efficiently a service or configuration item can be restore to service after a failure
  - ***Mean Time to Restore Service (MTRS)***

$$MTRS = \frac{Total\ downtime}{Number\ of\ breaks}$$

Availability Management



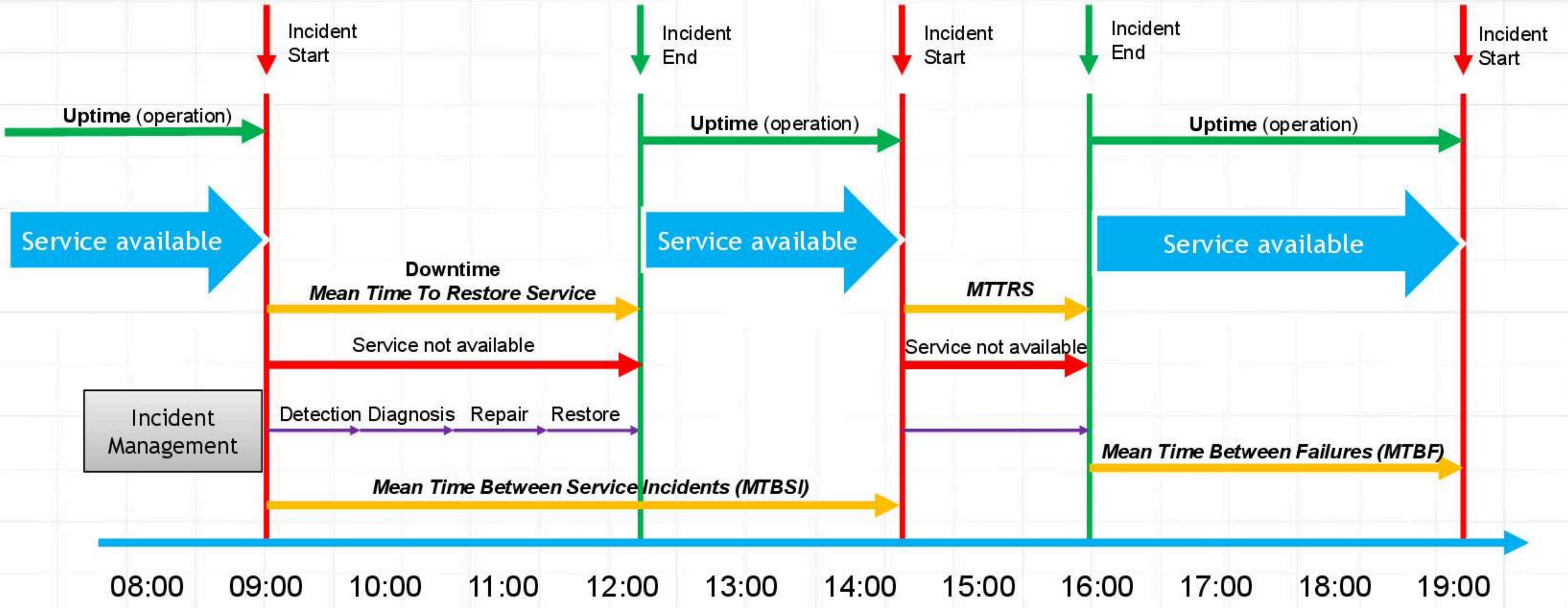
Availability Management



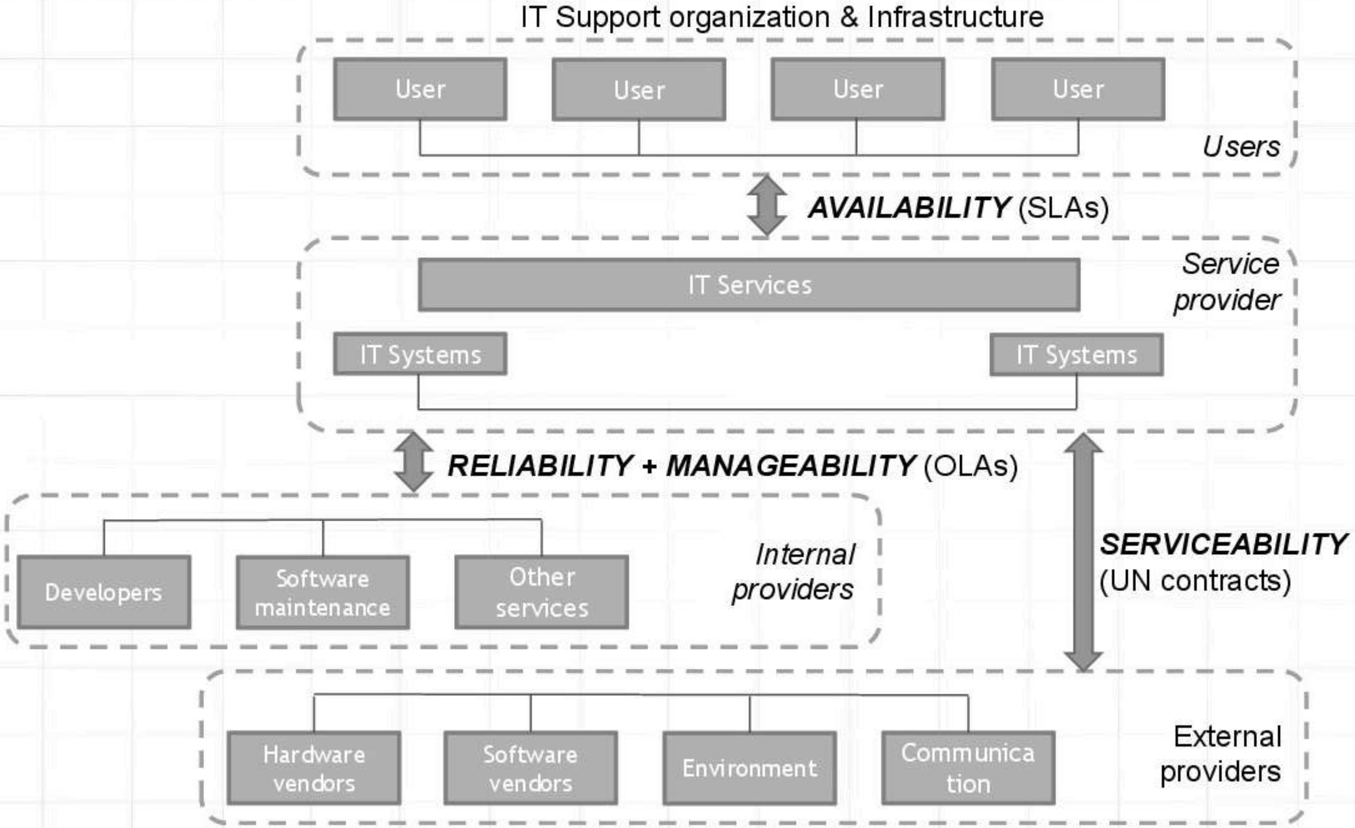
Availability Management

- ***Serviceability***
  - ability of the supplier or other entity to meet the terms of the contract
  - includes agreed levels of:
    - ***Availability***
    - ***Reliability***
    - ***Maintainability*** for a service or ancillary component

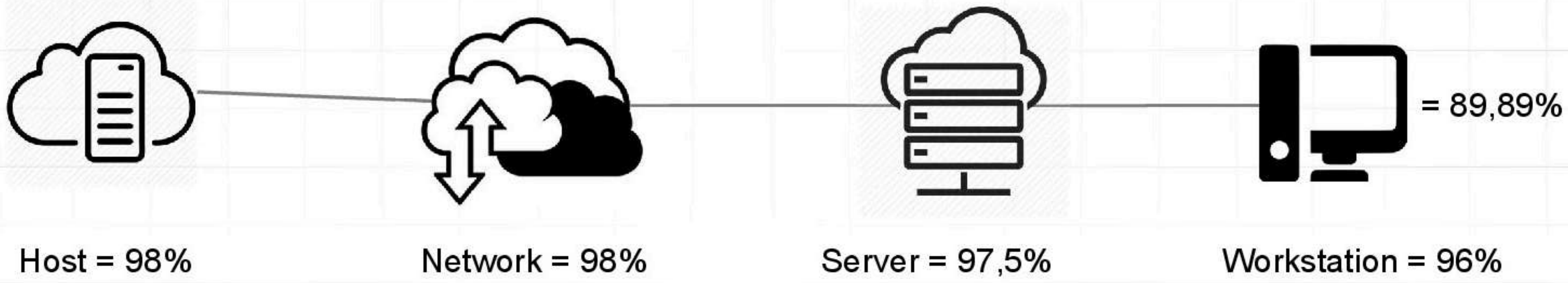
Availability Management



Availability Management



Calculation of availability levels



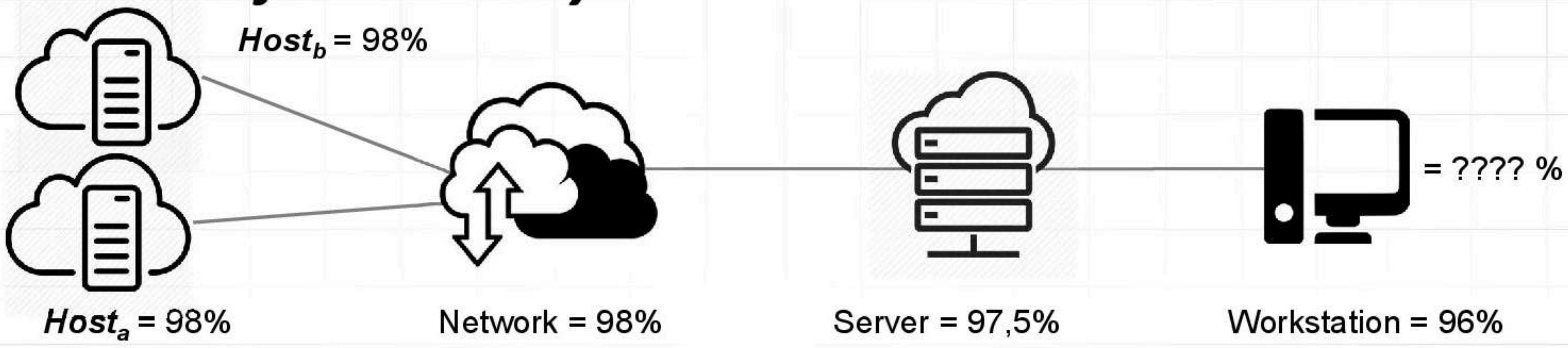
- ***Serial configuration***
  - Availability of the system from the user's workstation level is calculated from the formula:

$$A = A_H * A_N * A_S * A_W$$

  - *Which returns in result:*

$$A = 0{,}98 * 0{,}98 * 0{,}975 * 0{,}96 = \mathbf{0{,}8989}$$

Calculation of availability levels



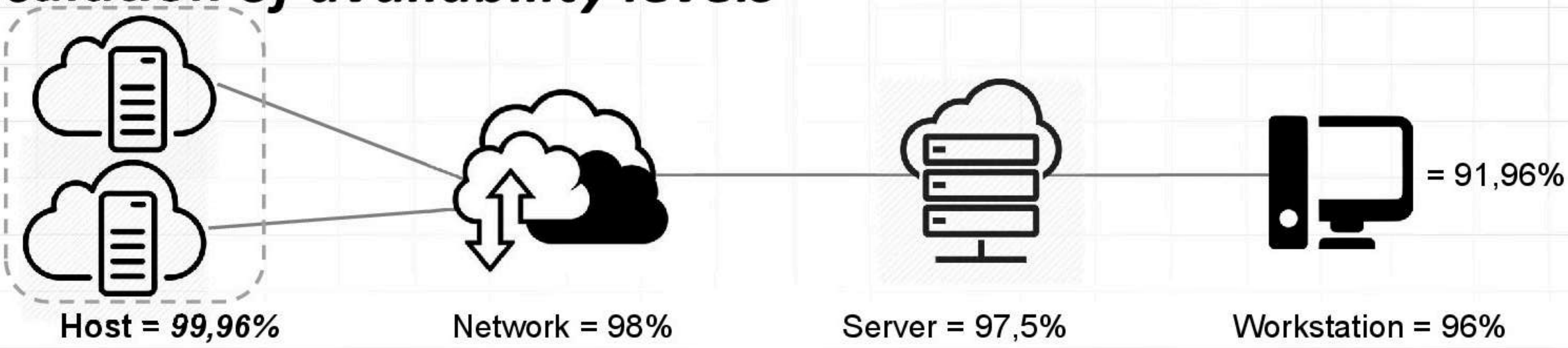
- ***Parallel configuration***
  - The host has been doubled to increases resilience. Host availability is calculated from the formula:

$$A = 1 - (1 - A_x)^2$$

  - *Which returns in result: :*

$$A_H = 1 - \left((1 - 0{,}98) * (1 - 0{,}98)\right) = 0{,}9996$$

Calculation of availability levels



- ***Parallel configuration***
  - Calculated host availability give the possibility to use the previous formula

$$A = A_H * A_N * A_S * A_W$$

  - *Which returns in result:*

$$A = 0{,}9996 * 0{,}98 * 0{,}975 * 0{,}96 = \mathbf{0{,}9196}$$

Availability expressed in nines

- How long downtime of a service will result in SLA breached?
- What percentage of the service's availability is expected by the recipient?
  - one 9s → 90 %
  - two 9s → 99 %
  - three 9s → 99,9 %
  - four 9s → 99,99 %
  - five 9s → 99,999 %

Availability [%]	Downtime / year	Downtime / month	Downtime / week	Availability class	Type of system
90	36 d 17 h 20 m	72 h	16,8 h	1	Unmanaged
95	18d 6h	36 h	8,4 h		
97	10,96 d	21,6 h	5,04 h		
98	7,30 d	14,4 h	3,36 h		
99	3 d 11 h 20 m	7,2 h	1,68 h	2	Managed
99,5	1,83 d	3,6 h	50,4 m		
99,8	17,52 h	86,23 m	20,16 m		
99,9	8,76 h	43,2 m	10,1 m	3	Well Managed
99,95	4,38 h	21,56 m	5,04 m		
99,99	52,56 m	4,32 m	1,01 m	4	Fault Tolerant
99,999	5,26 m	25,9 s	6,05 s	5	High-Availability
99,9999	31,5 s	2,59 s	0,605 s	6	Very-High-Availability
99,99999	3 s	0,25 s	0,004 s	7	Ultra-Availability

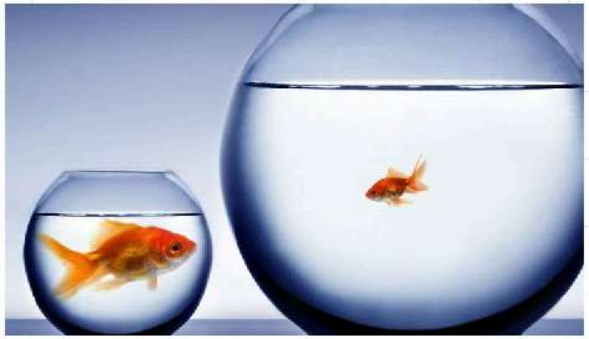
Cost of service unavailability (example)

- An IT service (A) with availability 99%), used by 1000 employees (U) with monthly salary (S) of 30 000 PLN,
- Total salary (P) 30 000 000 PLN / month, total cost (T_c) ➔

$$T_C = \left(U * S * \left(\frac{1 - A}{100}\right)\right) = 1000 * 30\,000 * 0{,}01 = 300\,000\ PLN$$

- Cost of 1h work in a month ➔ 30 000 000 / 160 h = 187 500 PLN
- Total number of man-hours of all employees in a month* ➔ r_g = U * 160 = 160 000
- Cost of unavailability of the service
  - 1h => cost 187 500 PLN
  - 1% of time => cost of 1,6 man-hour => 1,6 * 187 500 = 300 000 PLN

Capacity Management



- Capacity:
  - the ability of a system to operate at a given level of performance
- Process aims:
  - to monitor of resources to ensure the performance conditions described in the SLA
  - to plan for a possible change of resources to meet these conditions in the future
  - to strive to minimize infrastructure costs while maintaining the quality of services

IT Service Continuity Management

- Process Aim:
  - to plan and prepare the resources needed to ensure the continuity of the service in the event of a disaster, on the terms agreed in the SLA
- ***Disruption*** – an unplanned event that disrupts the service for a significant amount of time
- ***Disaster*** – an interruption of critical service activities for a significant part of the time

IT Service Continuity Management

- ***ITSCM*** counteracts and alleviates the effects :
  - loss, damage or refusal of access to key infrastructure
  - breakdown of key business services and applications
  - loss of performance (and failure) of services provided by suppliers, distributors or third parties
  - loss of key information or its misrepresentation
  - sabotage, extortion or commercial espionage
  - infiltration
  - attacks on critical information systems

Failover

- ability to automatically and seamlessly switch to a reliable system
- when the system / part of the system fails ➔ ***standby*** or ***redundancy*** should ensure failover and reduce / eliminate the negative impact on service level
- in order to achieve redundancy in case of abnormal failure or termination of the previously active version, the system or component must *always* be ready to automatically switch to operation

High availability systems

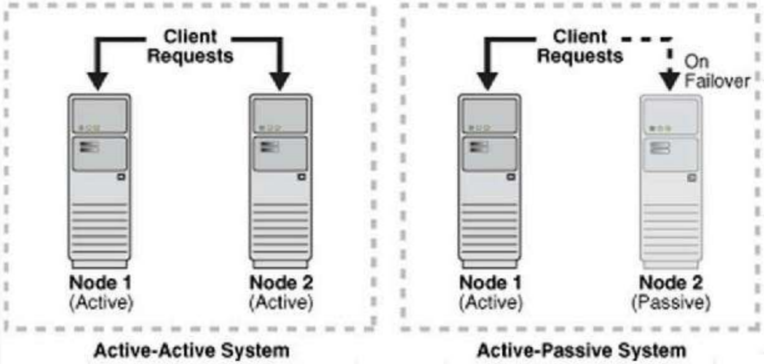
- ***High Availability*** - a feature of a system that is intended to provide an agreed-upon level of operational performance, usually uptime, for longer than normal.

- Mechanisms
  - redundancy
  - replication (synchronous / asynchronous)
  - cloning and mirroring
  - Load balancing
  - Failover Zones
  - clustering

- *Serverless compute*

High availability systems

- ***Active-Active***
  - uniform loading of all nodes
  - load balancing (preventing overloading of a single node)
  - improvement of throughput and response time
  - node configurations and settings should be identical



- ***Active-Passive (active-standby)***
  - there must be at least 2 nodes, 1 active
  - 2^{nd} node passive / on standby as emergency
  - 2^{nd} (backup) in standby mode in case the active (main) one stops working
  - if no failure occurs, clients only connect to the active server

https://docs.oracle.com/cd/B14099_19/core.1012/b14003/intro.htm

Failover

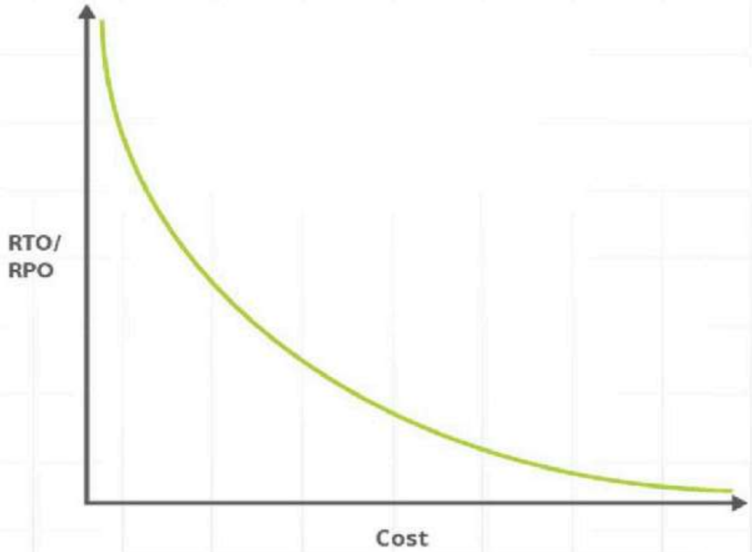
- ***Disaster Recovery***
  - processes, policies, and procedures related to resuming or maintaining information and communications infrastructure critical to the organization after a natural or man-made disaster
- ***Recovery Time Objective***
  - target duration and service level at which a business process must be restored after a failure (or disruption) to avoid unacceptable consequences associated with business continuity interruption
- ***Recovery Point Objective***
  - target point in time to which data will be irretrievably lost as a result of a major incident

Failover

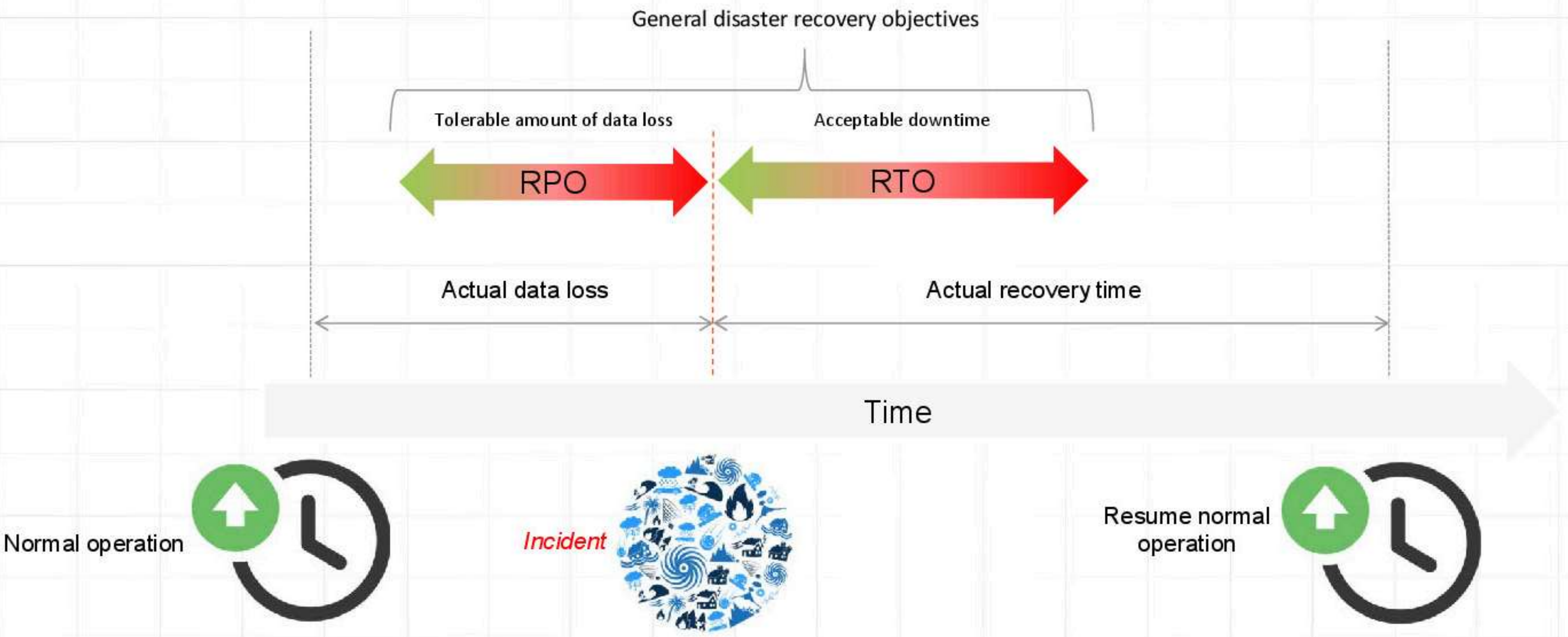
- ***Recovery Time Objective (RTO)***
  - defined by the SLA
  - maximum acceptable amount of time the system can be unavailable
- ***Recovery Point Objective (RPO)***
  - describes only the length of time (does not consider the quantity or quality of lost data)
  - point in time to which you want to restore the system
- RTO and RPO included in ***SLA***s

Failover

- Relationship of cost to RTO/RPO
  - smaller are RTO and RPO (quicker restore) higher cost



Disaster Recovery



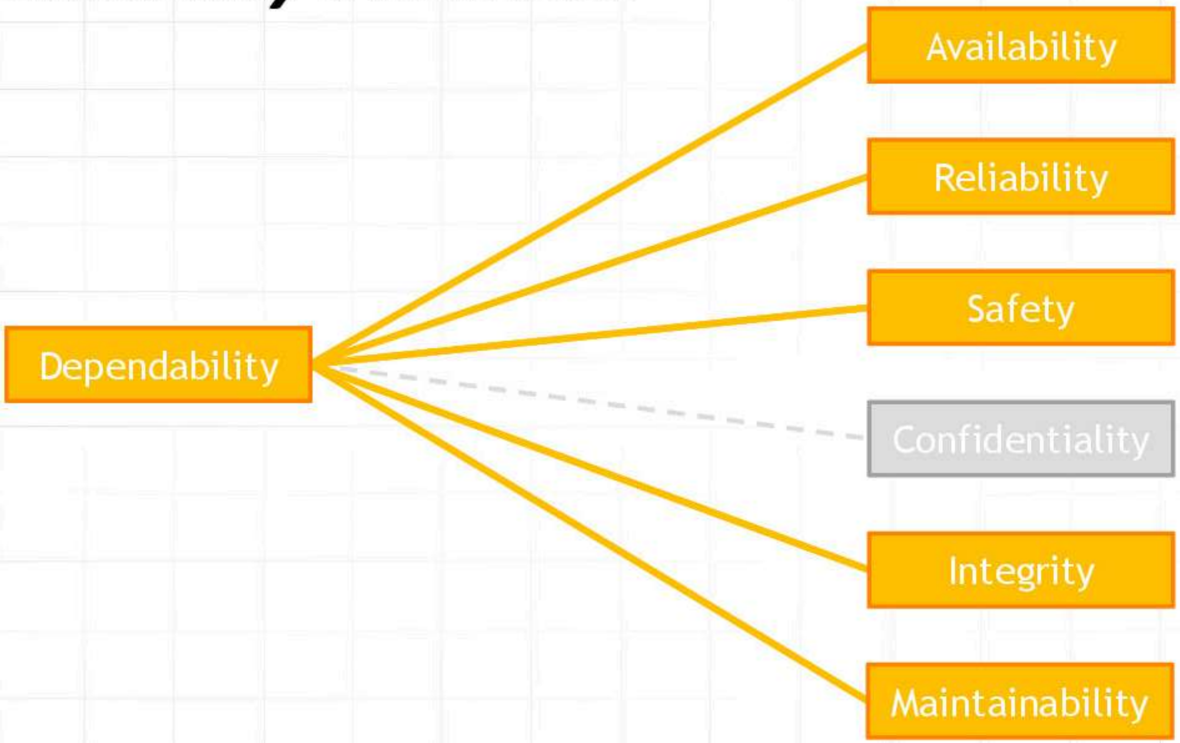
Restoring services to operation

Cold (Gradual Recovery)	• Servers must be configured, and applications installed ➔ time Days-Week(s) (> 72 h)
Warm (Intermediate Recovery)	• Configured servers, installed applications, connection to the production infrastructure ➔ time Minutes-Hours (max. 72 h)
Hot (Immediate /Fast Recovery)	• Full duplexing, redundancy - failover ➔ time (almost) IMMEDIATELY (max. 24 h)

Security vs. Safety

- ***Safety*** - freedom from unacceptable risk of physical injury or human health damage, directly or indirectly as a result of damage to property or the environment
- ***Safety*** is an attribute of ***dependability***

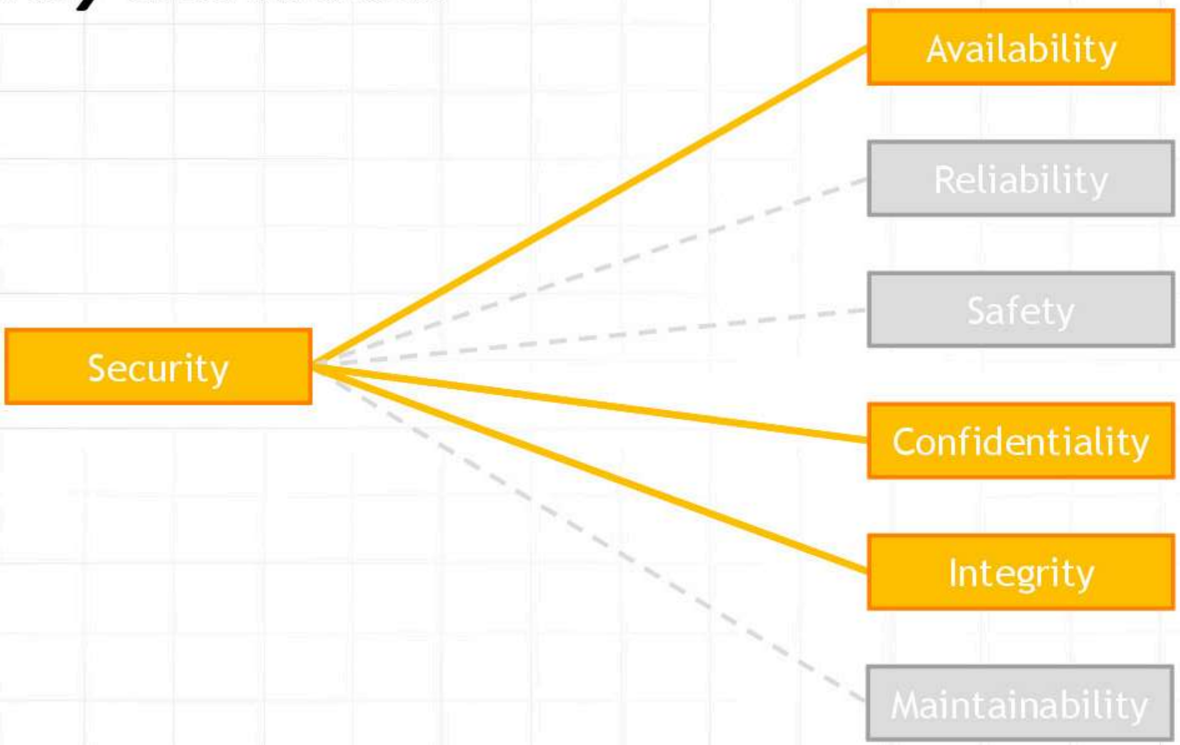
Dependability attributes



Security vs. Safety

- ***Security*** - prevention of illegal / unwanted penetration, intentional / unintentional interference in correct and intended operation or unauthorized access to confidential information in IT systems
- ***Security*** is a combination of:
  - ***availability***
  - ***confidentiality***
  - ***integrity***

Security attributes



Confidentiality

- ***Confidentiality***
  - is a security function
  - indicates the areas in which data should not be shared / disclosed to unauthorized persons, processes or other entities.

Integrity

- **Integrity**
  - property of data that excludes their modification in an unauthorized manner
  - inability to introduce an unauthorized change to the system

Authorization <> authentication

- ***authentication***
  - confirming the declared identity of the entity
- ***authorization***
  - process of granting rights to data to an entity